Laravel 13.x

Dokumentasi Lengkap Fitur Baru

Analisis Mendalam Changelog Resmi • September 2026 - Mei 2026

Laravel Framework 13.x • Scout • AI • Horizon • Inertia 3.x • Boost • MCP • Pao

Disusun untuk Project: laravel-inertia (Laravel 13 + Inertia v3 + React 19 + Wayfinder)

INFORMASI DOKUMEN	
Tanggal Analisis	14 Oktober 2026
Sumber Utama	https://laravel.com/framework/docs/changelog
Versi Laravel	13.x (Rilis Mei 2026) — The clean stack for Artisans and agents
Total Fitur Baru	41+ fitur utama (Framework, Scout, AI, Horizon, Inertia, dll.)
Fokus Rilis	AI-Native, Vector Search, DX, Performance

22	13	8
Framework Core	Scout + AI	Inertia 3.x

Daftar Isi

1. **Ringkasan Eksekutif** — Apa yang Baru di Laravel 13
2. September 2026 — 13 Fitur Terbaru (Vector, Storage, Collection, Testing)
3. Agustus 2026 — Cloud, Read-Through Filesystem, Debounce Queue, Scout Turbopuffer
4. Juni 2026 — Route Metadata, Bus::bulk(), artisan dev, Postgres Pooler
5. Mei 2026 — Rilis Laravel 13, Inertia 3.x, Laravel AI Sub-Agent, Pao, MCP
6. Tabel Rekomendasi Implementasi untuk Project Anda
7. Kesimpulan & Roadmap Adopsi

1. Ringkasan Eksekutif

Laravel 13 dirilis Mei 2026 dengan tema besar: AI-Native, Developer Experience, dan Performance. Berbeda dengan rilis sebelumnya, Laravel 13 menjadikan Vector Search, Semantic Search, dan AI Agent sebagai fitur inti framework, bukan package eksternal.

Tiga Pilar Utama:

- **AI & Vector:** AsVector cast, MariaDB/PostgreSQL vector queries, Scout semantic/hybrid untuk Typesense/Meilisearch/Database/Turbopuffer
- **Storage & Queue Performance:** copyToDisk/moveToDisk, read-through filesystem, Bus::bulk() bulk insert, Horizon log scaling
- **DX & Frontend:** php artisan dev (ganti composer dev), Route metadata, Inertia 3 poll/rescue, whenFilledEnum, chunkBy

Kategori	Jumlah	Dampak
Framework Core	22 fitur	Tinggi - Wajib dipahami
Scout (Search)	5 fitur	Tinggi jika ada search
Laravel AI	6 fitur	Tinggi untuk AI
Inertia 3.x	8 fitur	Wajib untuk project Inertia
Horizon / Queue	4 fitur	Sedang-Tinggi
Tooling (Pao/MCP/Boost)	6 fitur	Sedang

2. September 2026 — Update Terbaru (13 Fitur)

2.1 Semantic & Hybrid Search untuk Typesense (Scout)

Deskripsi: Engine Typesense mendukung pencarian semantik (makna) dan hybrid (keyword + makna) via semantic() dan hybrid(). Embedding bisa via Laravel AI toSearchableEmbedding(), auto-embedding native Typesense, atau precomputed vector.

```
Product::search('sepatu lari ringan')->hybrid()->get();  // gabung keyword + vector
```

```
Product::search('makna abstrak')->semantic()->get();  // hanya vector
```

Kasus: E-commerce (sepatu jogging ≈ running shoes), knowledge base natural language, RAG chatbot.

2.2 Reliable MariaDB Vector Queries + Cast AsVector

Deskripsi: Fix query vector MariaDB dengan vec_fromtext() + Cast AsVector handle binary MariaDB & text PostgreSQL sebagai array<float>.

```
protected $casts = ['embedding' => AsVector::class];
```

```
Document::query()->whereVectorSimilarTo('embedding', $vec)->orderByVectorDistance('embedding', $vec)->get();
```

Kasus: Aplikasi AI di MariaDB 11.7+ tanpa pindah pgvector. Simpan embedding OpenAI langsung.

2.3 Copy / Move File Antar Disk

```
Storage::disk('local')->copyToDisk('s3', 'reports/report.csv', 'archive/report.csv');
```

```
Storage::disk('local')->moveToDisk('backup', 'reports/report.csv');
```

Keunggulan: Streaming, auto close, move hapus source hanya jika sukses. Kasus: arsip harian local S3pindah upload tmp permanent.

2.4 Pilih Command di php artisan dev

```
DevCommands::withoutVendorCommands();  DevCommands::withoutDefaultCommands();
```

Kasus: Project dengan banyak package inject command ke dev, tapi hanya butuh server+vite+queue custom.

2.5 Mercure Broadcasting di Laravel Echo

Echo connector Mercure berbasis EventSource native, tanpa lib ekstra. Support private/presence/encrypted + whispers. Hub default /.well-known/mercure, config VITE_MERCURE_HUB_URL.

Kasus: Alternatif Pusher/Reverb self-host murah & compliant.

2.6 Chunk Collections by Adjacent Values (chunkBy)

```
collect([1,1,2,2,1])->chunkBy(fn($v)=>$v) // [[1,1],[2,2],[1]]
```

Beda groupBy: chunkBy hanya group yang berurutan. Support Collection & LazyCollection. Kasus: group log per sesi, pisah transaksi per hari berurutan, timeline chat per pengirim.

2.7 One-Time Mail & Notification Assertions

```
Mail::assertSentOnce(OrderShipped::class);  Notification::assertSentToOnce($user, InvoicePaid::class);
```

Alias pendek dari assertSentTimes(...,1). Kasus: Testing Pest pastikan OTP/invoice tidak double send.

2.8 Store SQL NULL dengan AsCollection::nullable()

```
protected $casts = ['items' => AsCollection::nullable()]; // simpan NULL native, bisa whereNull
```

Kasus: Field JSON optional yang perlu dibedakan kosong vs null untuk query & index.

2.9 Request In-Place MySQL Index (inplace)

```
Schema::table('video_sessions', fn($t)=> $t->index('video_id')->inplace()->lock('none')); // ALGORITHM=INPLACE
```

Jika tidak bisa MySQL reject (bukan silent fallback). Kasus: Migration tabel 10jt row tanpa lock lama.

2.10 Count Worker Crashes ke maxExceptions

#[CountCrashesAsExceptions] pada Job class. Worker mati OOM/kill dihitung ke maxExceptions agar tidak retry infinite. Tidak hitung normal complete & timeout.

2.11 AI: Build Provider dari Runtime Config

```
$agent->prompt($text, provider: [Ai::build(['driver'=>'anthropic','key'=>$tenant->ai_key])]);
```

Tanpa ubah global config. Support tenant-specific key & failover. Untuk queued agent build di provider() method. Kasus: SaaS multi-tenant.

2.12 Horizon: Logarithmic Auto-Scaling

Strategi baru 'log' selain 'time' & 'size'. Weight backlog logaritmik cegah 1 queue besar menyedot semua worker. Kasus: email(10k) vs notif(100) tetap adil.

2.13 Scout: Meilisearch Native Embedders

Set driver=>meilisearch + embedder, Meilisearch generate embedding. Bisa override options(['vector'=>...]). Kasus: AI search tanpa panggil OpenAI.

3. Agustus 2026 — Cloud, Filesystem & Queue

3.1 Laravel Cloud Facade

Cloud::hosted() | Cloud::usesManagedQueues() | Cloud::queue() — deteksi runtime di Laravel Cloud untuk switch connection. Kasus: bedakan config lokal vs Cloud.

3.2 Read-Through Filesystem

```
'assets' => ['driver'=>'read-through','primary'=>'r2','fallback'=>'legacy-s3']
```

Cek primary dulu, jika tidak ada ambil fallback & auto-copy ke primary. Write tetap primary. Support streamed read. Kasus: migrasi S3 lama R2 tanpa downtime, file lama tercopy otomatis saat diakses.

3.3 Portable AsVector & MariaDB Vector Queries

Laravel support vector di PG & MariaDB API identik: whereVectorSimilarTo, orderByVectorDistance. Kasus: RAG tanpa vector DB eksternal, cukup kolom vector(1536).

3.4 Auto Retry Safe Redis Commands

PhpRedis auto reconnect & retry read-only + SET 1x jika transient fail. Config REDIS_COMMAND_RETRIES. Kasus: Redis di k8s reconnect sepersekian detik.

3.5 Debounce Queued Event Listeners

```
#[DebounceFor(30, maxWait:120)] class UpdateProductSearchIndex implements ShouldQueue { debounceId() { return (string)$event->product->id; } }
```

Jika 10 event sama dalam 30s hanya proses terakhir. Kasus: update search index setelah burst edit, sync Algolia.

3.6 Pause Semua Queue Sekaligus

php artisan queue:pause --all & resume --all + event QueuesPaused/Resumed. Kasus: maintenance/deploy global pause.

3.7 Scout: Turbopuffer & Database Semantic Search

```
Document::search('ocean climate')->semantic()->where('published', true)->get();
```

Engine Turbopuffer tanpa SDK + Database engine support semantic/hybrid murni pakai DB. Kasus: startup tanpa budget Meilisearch, AI search hanya MySQL/Postgres + Laravel AI.

3.8 Scout: Semantic & Hybrid Meilisearch

Article::search('guide queues')->hybrid(textWeight:1, semanticWeight:2)->get(); — via Laravel AI embedder.

3.9 AI: Repair Unknown Tool Calls

#[RepairToolCalls] — jika LLM panggil tool tidak ada, return daftar tools agar self-repair, bukan abort. Kasus: agent banyak tools typo nama.

4. Juni 2026 — DX & Performance Massal

4.1 Route Metadata

Route::metadata(['head'=>['title'=>'Users']])->group(...); $request->route()->getMetadata('head.title'); — merge recursive & route cache. Kasus: SEO title, feature flag, permission tag.

4.2 Postgres Transaction Pooler

pooled=>true untuk PgBouncer/RDS Proxy/Neon. Handle emulated prepares. DDL pakai ::direct. Kasus: Supabase/Neon tanpa error prepared statement.

4.3 Should Not Retry Exception

retry() return false di Exception atau withExceptions() di bootstrap/app.php. Kasus: validasi gagal, payment declined jangan habiskan attempt.

4.4 schema:dump --without-migration-data

Dump struktur tanpa migrations table. Kasus: test env seed fresh.

4.5 Fix between()/unlessBetween() Timezone

Dulu between() sebelum timezone() pakai UTC diam-diam, sekarang deferred. Kasus: schedule Asia/Jakarta yang sebelumnya jam salah.

4.6 php artisan dev (Pengganti composer dev)

Server+queue+log+Vite concurrent color-coded, auto detect npm/yarn/pnpm/bun. DevCommands::artisan('reverb:start')->orange(). Kasus: 1 command dev lengkap.

4.7 array Maintenance Mode Driver

In-memory per process hindari race file saat parallel testing. Kasus: php artisan test --parallel tidak flaky.

4.8 whenFilledEnum()

$request->whenFilledEnum('status', Status::class, fn(Status $s)=> $query->where('status',$s)); — collapse tryFrom+null guard. Kasus: filter enum tanpa boilerplate.

4.9 anyOf & Multi-Type Union JsonSchema

Untuk AI structured output OpenAI/Gemini field bisa string|object|array. JsonSchema::union(['string','number'])->nullable(); Kasus: agent JSON fleksibel.

4.10 Queue Attributes di Traits

#[Queue(connection:'redis')] di trait otomatis kebawa Job. Kasus: 20 Job share config sama di 1 trait.

4.11 Bus::bulk() — Bulk Insert Efisien

Bus::bulk($users->map(fn($u)=>new ProcessUser($u))->all()); — 1 bulk insert per group queue/connection. Kasus: 50k email, 100k import tanpa overhead.

4.12 attachFromStorage di Notification

(new MailMessage)->attachFromStorage('invoices/1.pdf'); Kasus: kirim invoice PDF dari S3 via Notification.

4.13 Cache rememberWithState & Str::studly normalize

Cache::rememberWithState() return [value, hitBool] untuk metrics. Str::studly('CBOR', normalize:true)=>'Cbor' untuk ALL_CAPS.

5. Mei 2026 — Rilis Laravel 13 & Inertia 3.x

5.1 Laravel 13 Skeleton Baru

- nunomaduro/pao included default — structured AI agent output

- api/* otomatis JSON tanpa header Accept via shouldRenderJsonWhen()
- Vite Font Plugin di starter kits, passwordrules attribute untuk password manager

5.2 Inertia 3.x — Wajib untuk Project Anda

`router.poll(2000,{},{mode:"cancel"}) // abort in-flight | mode:"rest" tunggu interval | mode:"overlap" lama`

usePoll(10_000, () => ({only:["notifications"], data:{since}})) — re-evaluate tiap tick agar tidak stale. router.once("before",...) — listener sekali pakai.

`Deferred rescue: <Deferred data="stats"><template #rescue="{reloading}"><button @click="router.reload({only:['stats']})">Retry</button></template></Deferred>`

CSP nonce di createInertiaApp({nonce:"..."}), sandbox iframe, SSR host option. Kasus: polling notifikasi tanpa dobel, props berat dengan retry, CSP ketat.

5.3 Laravel AI: Sub-Agent as Tools

Parent return [new WebSearch, new ResearchAgent] di tools(), sub-agent ActsAsTool isolated context. Plus HasConversations trait, stream_failed broadcast, Str::toAudio() TTS, failover streaming fix.

Kasus: ProjectManager delegasi ke ResearchAgent untuk riset mendalam.

5.4 Boost, MCP, Pao, Moat

- **Boost:** Tinker MCP opt-in, hapus model discovery broken, sync Flux UI
- **MCP:** ResourceLink, Icon attribute (spec 2025-06/11-25)
- **Pao:** Rector support, PHPStan JSON, PAO_FORCE=1 untuk Docker
- **Moat:** Scan security posture GitHub (2FA, branch protection, secret scanning)
- **Installer:** laravel new output JSON jika dijalankan AI agent

6. Tabel Rekomendasi Implementasi

Prioritas adopsi untuk project laravel-inertia (Laravel 13 + Inertia v3):

Kebutuhan	Fitur Baru	Prioritas
Search pintar produk/dokumen	Scout hybrid() + AsVector	Tinggi
Migrasi storage S3 lama	read-through driver	Tinggi
Arsip file antar disk	copyToDisk / moveToDisk	Tinggi
Queue 10k+ job	Bus::bulk()	Tinggi
Update index anti-burst	#[DebounceFor]	Sedang
Deploy tanpa ganggu	queue:pause --all	Sedang
Filter status enum	whenFilledEnum()	Tinggi
SEO per route	Route::metadata()	Sedang
Dev 1 command	php artisan dev	Tinggi
Polling realtime	router.poll cancel	Tinggi
Props berat + retry	Deferred rescue	Tinggi
Multi-tenant AI key	Ai::build()	Jika AI

Kebutuhan	Fitur Baru	Prioritas
Tabel besar migration	inplace() + lock	Sedang

7. Kesimpulan & Roadmap Adopsi

Laravel 13 adalah rilis AI-native pertama. Jika belum butuh AI, fokus pada DX (artisan dev, whenFilledEnum, Route metadata) dan performance (Bus::bulk, copyToDisk, read-through, Horizon log). Jika ada roadmap search/chatbot, AsVector + Scout hybrid adalah fondasi paling murah tanpa vector DB eksternal.

1. Minggu 1: Ganti composer dev php artisan dev, adopt whenFilledEnum & chunkBy
2. Minggu 2: Migrasi file handling ke copyToDisk/moveToDisk, evaluasi read-through untuk migrasi storage
3. Minggu 3: Optimasi queue dengan Bus::bulk() & DebounceFor jika ada burst
4. Minggu 4: Inertia 3 poll mode & Deferred rescue untuk UX dashboard
5. Bulan 2: PoC Scout hybrid search dengan Database engine (tanpa Meilisearch) + AsVector

— Dokumen dihasilkan otomatis dari changelog resmi Laravel —

Sumber: https://laravel.com/framework/docs/changelog • Dibuat: 14 Oktober 2026